

Will **LITHIUM-AIR BATTERIES** Replace Lithium-Ion Batteries In Electric Vehicles Soon?

The extremely high theoretical energy density of lithium-air batteries encourages researchers to develop commercially viable batteries. However, the gap between idea and reality is still big. The challenge lies in improving the cycling stability under complex superoxide/peroxide containing situations while keeping a good energy efficiency and rate capability. The batteries are undergoing high-speed development and admirable breakthrough in improving their electrochemical performance



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The lithium-air battery (LAB) is a metal-air electrochemical cell which during discharge oxidises lithium metal at the anode and reduces oxygen from air at the cathode to induce a current flow. The process can be reversed by applying an external potential and such a battery can be electrically recharged. Because oxygen can be supplied continuously from air rather than stored in finite amounts within the cell, lithium-air batteries can theoretically provide an energy density ten times that of conventional lithium-ion batteries. But there are hurdles to overcome.

In the course of last decade, great attention has been devoted by many top academic and industrial laboratories worldwide, which now has yielded solutions to some of the hurdles. To make lithium-air battery technology commercially viable, two goals are highly important: a suitable cathode that uses a highly active and stable catalyst to enhance oxygen reduction reaction (ORR) and oxygen evolution reaction (OER) kinetics and a proper electrolyte design.

In the quest for finding a highly active catalyst, scientists have developed efficient catalysts like nanolithia, trimolybdenum phosphide, lithium iodide, etc. New electrolyte formulations have been developed that significantly reduce side reactions in the battery chemistry to enable longer cycle stability. Improved air-cathode structure with nanocarbon-containing catalyst increases the practical energy density.

With all these improvements the resulting battery works in the laboratory with high energy efficiency of 90%, an energy density of about 1500Wh/kg (about eight times better than Li-ion battery), and a long cycle life of 1000. PolyPlus Battery Company is now planning to launch world's first commercial lithium-air battery in the USA.

One of the bottlenecks in widespread implementation of sustainable energy technologies is a highly-efficient energy storage system. Lithium-ion batteries with energy capacity of 100 to 265Wh/kg are the prevailing solution for today's electrical and electronic devices. Various research studies now show that lithium-air batteries have the potential to reach an estimated energy density of 1300Wh/kg in future and about 500Wh/kg immediately in sight.

Energy density

Lithium-air batteries are often considered the ultimate batteries due to their theoretical energy density, which is many times more than of traditional lithium-ion batteries, making them comparable to gasoline. This is because the batteries utilise ambient oxygen and most of the cell volume is occupied by the anode while the cathode-active material is not stored in the battery. Lithium metal is a tempting anode material for any battery because of its outstanding specific capacity (3842mAh per gram against 815mAh per gm for zinc).

Lithium-air batteries have the potential